

# Course Wrap-Up & Capstone Project Presentations

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Date: 30th March 2026

# Today's Objectives

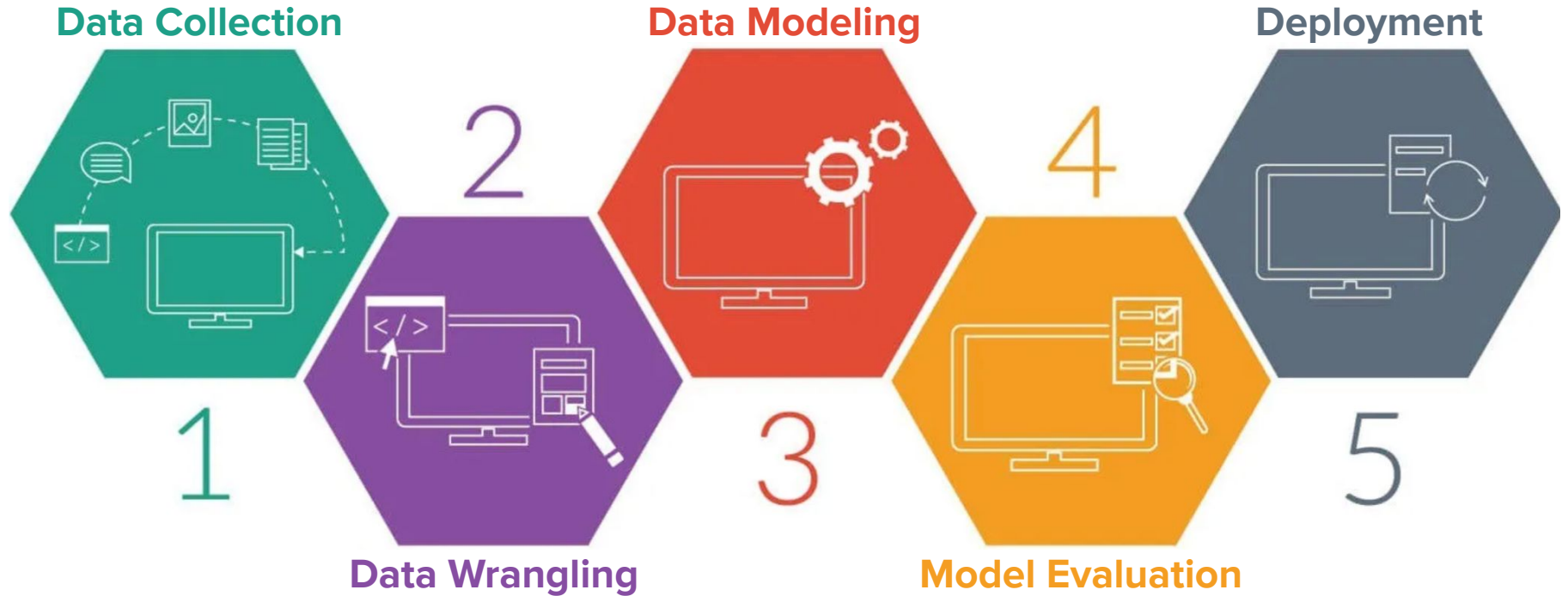
- **Review** key course content:
  - ML & AI in Clinical Bioinformatics
  - Data Preprocessing & Cleaning
  - Supervised Learning
  - Unsupervised Learning
  - Applications:
    - Survival Analysis
    - Radiomics
    - Pharmacogenomics
    - Radiomics
  - Bias, Fairness and Open Science
- **Final Project Overview**
- Course Evaluation Forms



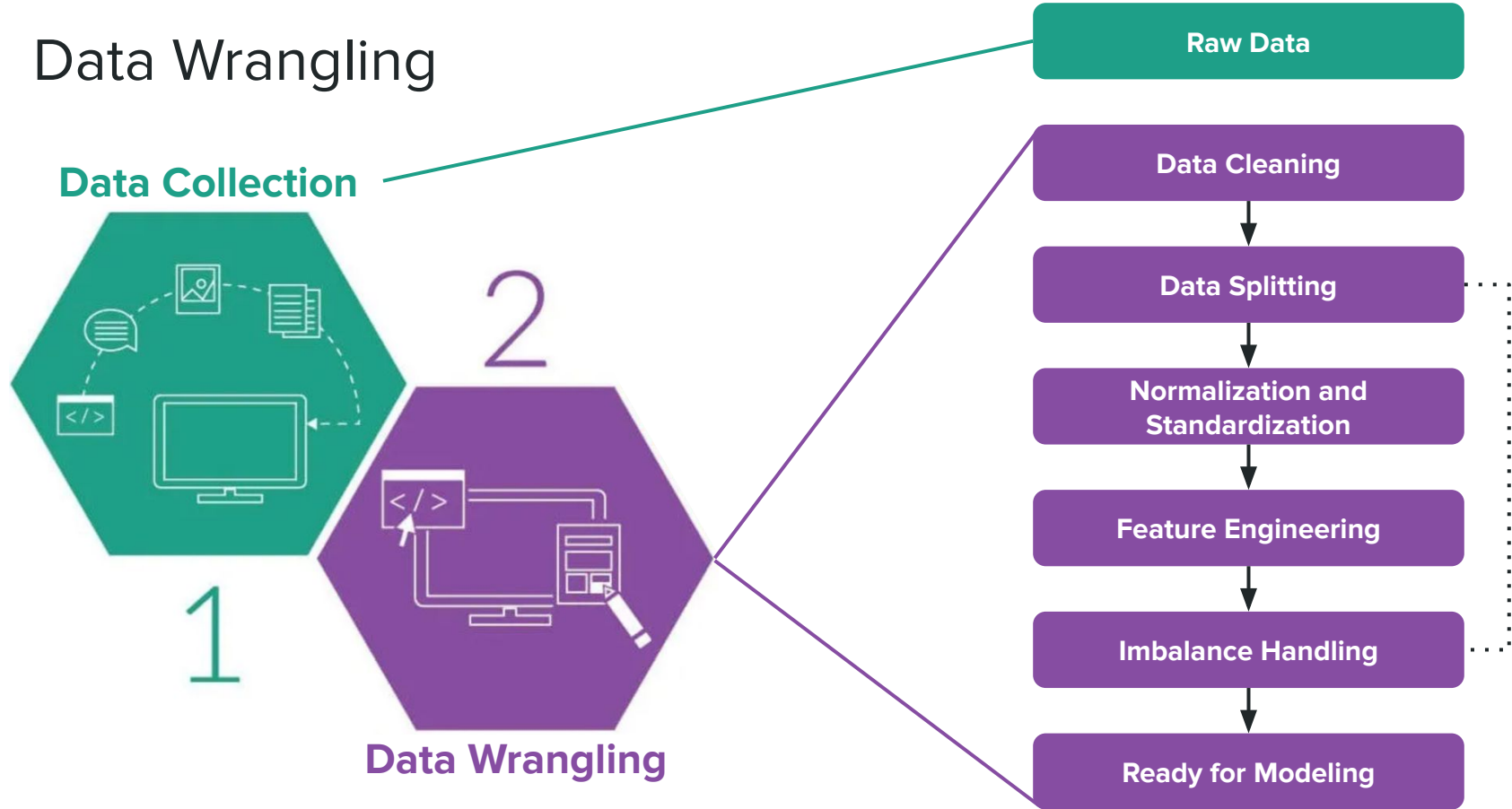
# Course Review

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# ML & AI in Clinical Bioinformatics



# Data Wrangling



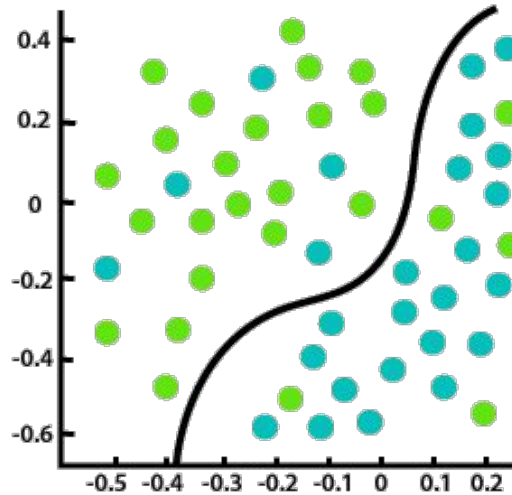
# Supervised Learning

- **Regression:**

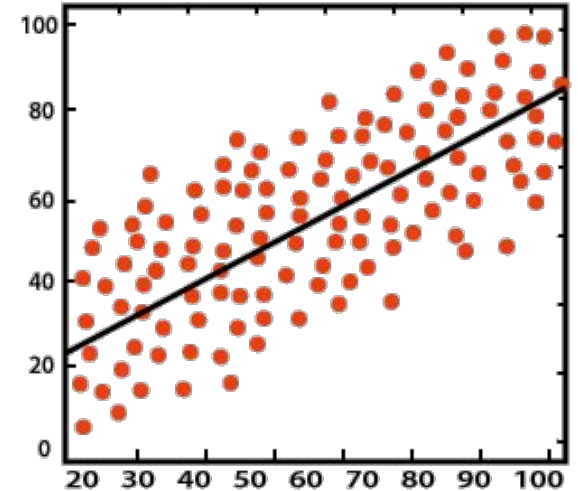
- Linear
- Polynomial

- **Classification:**

- Logistic regression
- k-Nearest Neighbours
- Decision Trees
- Random Forests
- SVM



Classification



Regression

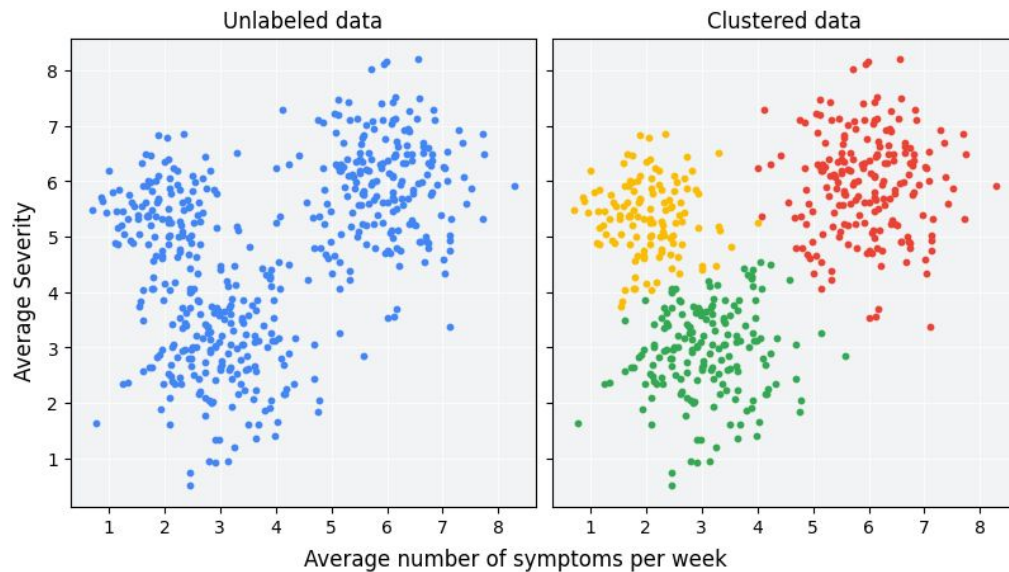
# Unsupervised Learning

- **Clustering:**

- k-Means
- Hierarchical
- DBSCAN\*\*

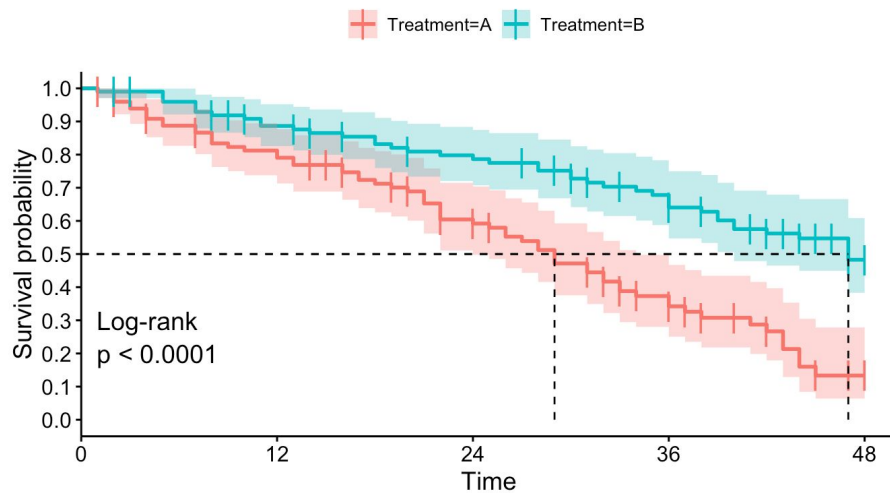
- **Dimensionality Reduction:**

- PCA
- UMAP
- t-SNE

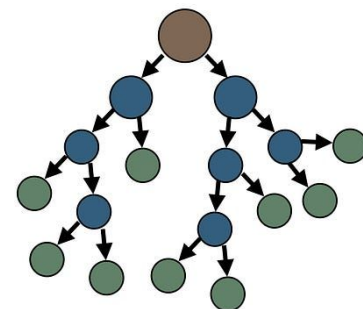


# Survival Analysis

- Cox Proportional Hazards;
- Random Survival Forests;
- Kaplan-Meier Analysis

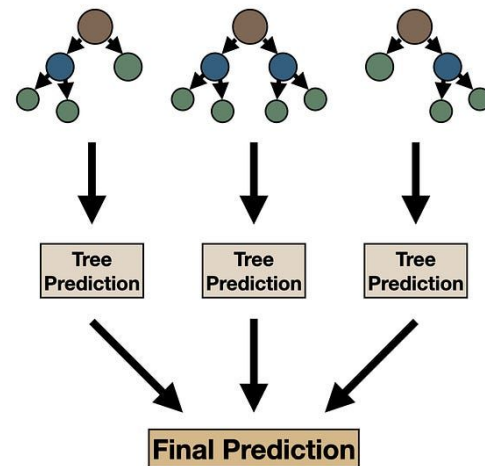


Decision  
Tree



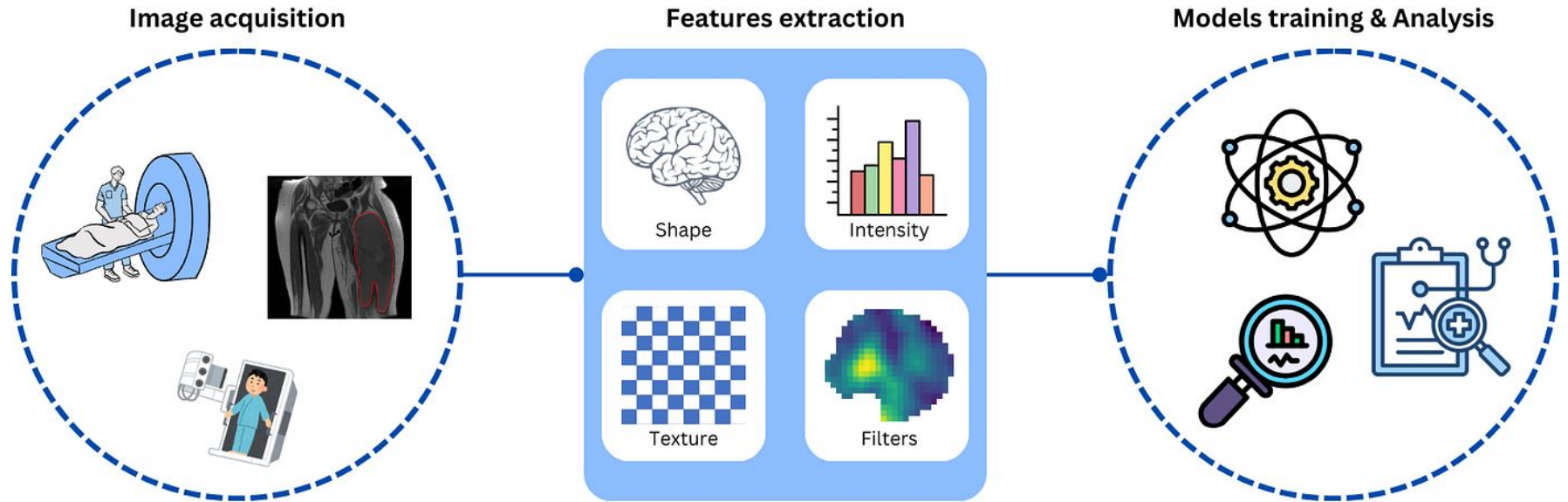
Final Prediction

Random  
Forest



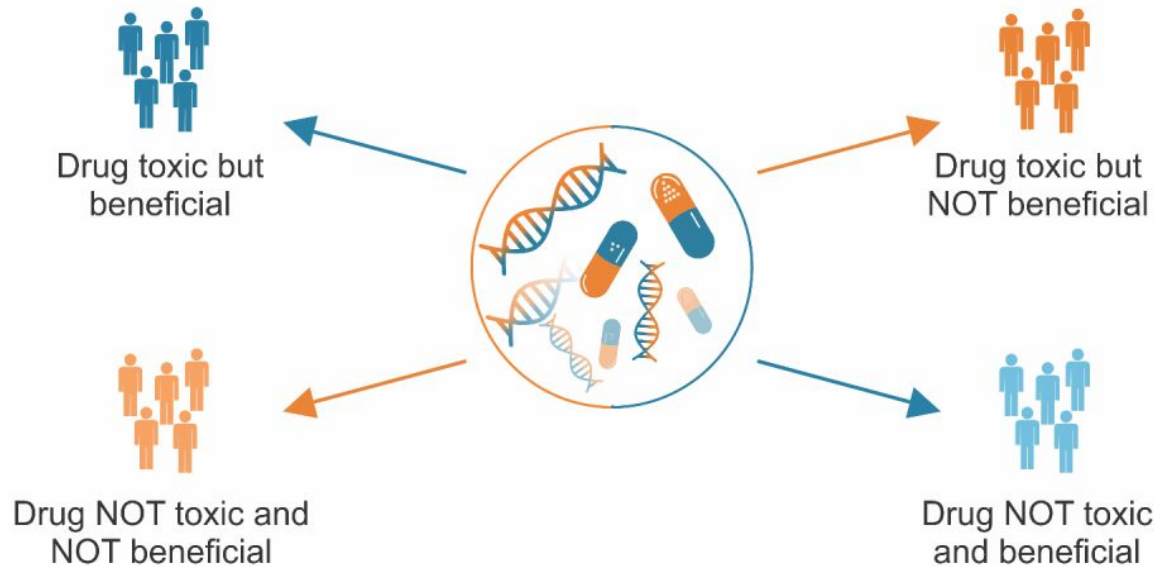
# Radiomics

- Extraction of quantitative imaging features to uncover hidden patterns.



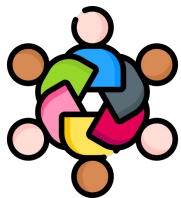
# Pharmacogenomics

- Study of how genetic variation affects an individual's response to drugs;
- **Role in precision medicine** → Tailored treatment plans.



# Bias, Fairness and Open Science

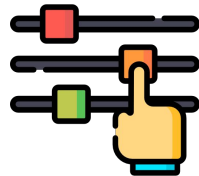
- **Sources of bias:** dataset, measurement, algorithmic
- **Bias mitigation strategies:**



Diverse & Representative  
Datasets



Bias Auditing &  
Fairness Metrics



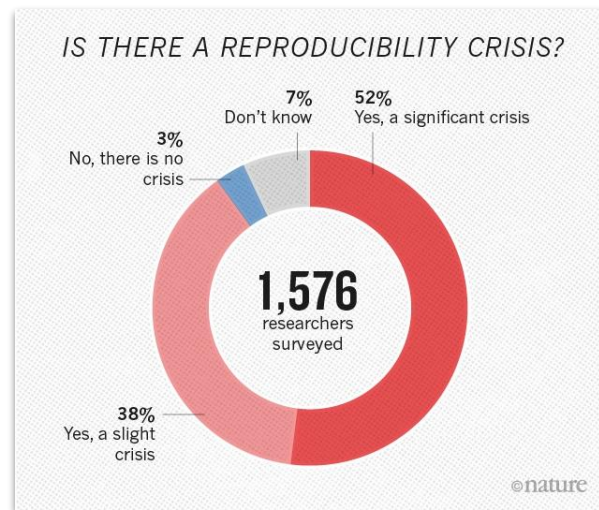
Algorithmic  
Adjustments



Transparency & Explainability



Open Science & Collaboration



|          |           | Data         |               |
|----------|-----------|--------------|---------------|
|          |           | Same         | Different     |
| Analysis | Same      | Reproducible | Replicable    |
|          | Different | Robust       | Generalizable |

# Final Project Overview

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# Grading Breakdown

| Section                          | Percentage |
|----------------------------------|------------|
| <b>Written Report (75%)</b>      |            |
| Introduction                     | 15%        |
| How It Works                     | 20%        |
| Use Cases & Performance          | 15%        |
| Code Implementation              | 15%        |
| Comparison                       | 20%        |
| <b>Presentation (25%)</b>        |            |
| Clarity & Organization           | 10%        |
| Key Insights from Implementation | 10%        |
| Visualizations or Live Demo      | 5%         |

## 1.1 Introduction

- Provide an overview of the chosen algorithm and explain its significance.
- Discuss why it is relevant to machine learning and how it differs from algorithms covered in class.
- Highlight its applications in real-world scenarios.

## 1.2 How It Works

- Explain the mechanics of the algorithm, including its fundamental principles.
- If applicable, include mathematical formulations and key parameters.
- Provide a high-level intuition behind the algorithm's decision-making process.

## 1.3 Use Cases & Performance

- Discuss when and where the algorithm works best.
- Identify its advantages and limitations compared to similar techniques.
- Provide real-world examples of its application.

## 1.4 Code Implementation

- Include a short Python snippet demonstrating the algorithm in action.
- Clearly explain the key components of the code, including data preprocessing steps, parameter tuning, and output interpretation.
- Ensure the code is well-documented with comments.

## 1.5 Comparison / Visualization

- Select a dataset and compare the research algorithm to a related algorithm covered in class (where applicable; for time-series algorithms, you could consider comparing different subsets of data versus comparing different algorithms).
- Provide visualizations (e.g., AUROC for classification, clustering plots, error distributions) to showcase performance differences.
- Explain any data transformations, assumptions, and key insights from the comparison.

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Introduction

How it Works

Use Cases &  
Performance

Code  
Implementation

Comparison /  
Visualization

## 2.1 Clarity & Organization

- Present a well-structured, concise explanation of the algorithm.
- Ensure smooth transitions between topics.

## 2.2 Key Insights from Implementation

- Highlight key findings from your implementation and comparison.
- Discuss strengths and limitations based on your results.

## 2.3 Visualizations or Live Demonstration

- Use relevant visualizations or a brief live code demo to illustrate your findings.
- Ensure visuals are easy to interpret and enhance understanding.